

Vexa: Challenges, What to Remove, and the Path to a Truly Personal Assistant

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The goal

Vexa should be a personal assistant that genuinely knows its user and cuts the repetitive work in their day. The value is not in tapping buttons; it is in an assistant that understands one person's world and acts on it.

The working one-liner: **Vexa is the private, local-first assistant that learns you and never ships your life to someone else's server.**

Two things it must get right:

- **Personalisation:** memory that grows from every conversation, so the experience feels made for that one person.
- **Removing repetition:** take over the daily, repeated thinking work, while leaving the parts of life people enjoy alone (calls with family, games, casual scrolling).

Where Vexa is today

Vexa today is an Android voice assistant that drives the phone through accessibility services, with AI choosing what to tap.

1. The user gives a voice instruction.
2. Vexa turns on the accessibility service.
3. It reads the screen and asks the AI which element to click.
4. The accessibility service performs the click, then repeats for the next step.

Supporting pieces already built:

- A multi-agent setup built with Claude Code: a **planner agent**, a **learning agent** and a **verification agent**.
- Google's open knowledge format (App Actions vocabulary) for describing actions.
- A hosted backend, Vexa Brain, on Render, with per-chat context-window management.
- A free-model fallback chain: when one free model's tokens run out, it falls through to OpenRouter models, then Groq models, so no model costs are paid.
- LangSmith tracing for debugging.
- A self-improving memory that grows from the user's conversations.

Challenges discussed

Eight problems came up; the first four are technical, the last four are about whether people will use it.

#	Challenge	What happens	Root cause
1	Automations fail midway	"Open WhatsApp and reply to Vivek based on his last message" opened WhatsApp but sent the wrong message	Context from an earlier screen (Vivek's message) is lost by the compose step; the accessibility tree alone is often ambiguous
2	Screenshots are too costly	Vision input would fix element-picking, but every step would burn paid tokens	Free-model fallback chain cannot afford images on each step
3	Banking and sensitive apps block Vexa	Bank apps refuse to open when any accessibility service is on	Google and apps are tightening accessibility use because scammers abuse it
4	Email feature broke on the server	SMTP calls inside the FastAPI HTTPS server threw errors every time	Wrong protocol: SMTP / EmailJS / Nodemailer used for reading and sending
5	No clear difference from ChatGPT or Claude	Their connectors already summarise Gmail and run scheduled tasks	Vexa was competing on features big companies already have
6	Privacy is not fully real	Data still goes to third-party cloud LLMs, even with no own server	Running on cloud models means text leaves the phone
7	Local storage will grow	Two to three years of use could fill the phone	Storing everything instead of a distilled memory
8	No target user	"Everyone" means no one opens it over ChatGPT	No specific person or painful problem chosen yet

What to remove from Vexa

Remove the accessibility service completely. It is the root of the reliability, trust and bank-app problems, and the platform is shrinking under it.

Remove	Replace with
Accessibility-service app control	Android's open doors: notification access, share sheet, intents and deep links
Screen-tapping automation	Vexa does the thinking (drafts, summaries, reminders); the user taps send
SMTP, EmailJS, Nodemailer	Gmail API over HTTPS with OAuth (reads and sends)
Website scraping for news	RSS feeds or a free news API tier
Storing every message	A small distilled memory of facts about the user
Any handling of bank or payment data	Nothing; "Vexa doesn't handle your money" is a feature
The "for everyone" pitch	One specific user group first

The new promise: **Vexa doesn't tap your screen; it does the thinking, and you stay in control.**

The honest question: why not just ChatGPT or Claude?

Right now Vexa has no feature the big assistants lack; email summaries, memory and scheduled tasks all exist there already. Competing on features against billion-dollar companies is a losing game.

Where a small builder can win is depth, not breadth:

- **They are a mile wide and an inch deep about you.** Vexa can understand one kind of person's world far better than a general tool.
- **Channels they ignore.** In India, business and daily life run on WhatsApp; most Western tools treat it as an afterthought.
- **Trust by design.** A tight scope (no bank data, minimal memory, honest data handling) is easier to trust than an assistant that wants everything.

The email briefing is not the moat. It is only what makes Vexa useful on day one.

Privacy, storage and trust

Don't go fully local and don't claim to. Use a hybrid model and be honest about it.

Storage that stays small for years

- Store distilled facts about the user ("works at Cognizant", "prefers short replies"), not every message.
- Merge and expire old facts regularly, so memory stays in the kilobytes-to-low-megabytes range even after years.
- Keep raw data (emails, chats) in its source app and fetch it when needed instead of copying it.

Being honest about cloud models

- Keep the memory on the device.
- Send the cloud model only the minimum text one task needs, never the whole memory.
- Never store request content on Vexa's own servers.
- Prefer providers that do not train on API data, and let privacy-minded users plug in their own API key.
- Say it plainly in the app: "We don't store your data; it goes to the model only for the moment it's needed."

Hard lines

- Vexa never reads, stores or acts on bank, payment or OTP data.
- The user can see and delete anything Vexa remembers about them.

The new direction: personalisation as the product

Reframe Vexa from "it does things in your apps" to "it does the thinking work you repeat every day, because it knows you." Most daily repetition is mental: scanning, remembering, drafting and following up.

How the personal experience works

1. **It learns you.** Memory grows from chats: your role, people, tone, goals and routines.
2. **It watches the right inputs.** Gmail, notifications and things you share into it, all through official APIs.
3. **It prepares, you decide.** Vexa drafts the reply, summary or follow-up in your voice; you approve or send.
4. **It gets better.** Every correction and approval sharpens the memory, so the next draft needs fewer edits.

Candidate directions explored

Direction	Who it serves	Assessment
Daily briefing (email + tech news)	Anyone who checks mail and news every morning	Easy first win, but ChatGPT and Claude already do it; a starting feature, not the product
Fresher job-hunt assistant	Students and freshers tracking applications	Rejected by you as not exciting enough
Outreach assistant for solo operators	Freelancers, founders, small agencies doing their own sales	Strongest so far: painful repetitive work, people pay, and WhatsApp is an edge

Outreach assistant in detail

- Knows the user's pitch, profile and offer from memory.
- Drafts personalised follow-ups in their voice across email and WhatsApp.
- Tracks who replied, who went cold and who needs a nudge today.
- Caution: use the official WhatsApp Business API; unofficial automation gets numbers banned. It is also a crowded sales-tool market, so it must be clearly better for a specific slice.

First feature and build sequence

Start with the daily 8 a.m. briefing: email summary plus tech news, delivered by email and push notification, running on the cloud even when the phone is off. Each step works on its own and builds on the last.

1. **Gmail access.** Create a Google Cloud project, enable the Gmail API, add OAuth so the user grants read and send once. Test by fetching today's emails.
2. **Email summariser.** Send fetched emails through the existing free-model fallback chain with a prompt that separates what matters from noise.
3. **Email search.** Bring back "any emails about Cognizant?" as a Gmail API search query, then summarise the results.
4. **News digest.** Pull five or six tech RSS feeds, filter by the user's interests from memory, and summarise.
5. **Schedule and deliver.** A cron job on the server at 8 a.m. runs steps 1 to 4, builds one report, emails it via the Gmail API and sends a push notification.

If the outreach direction is chosen, the same Gmail and memory pieces become its foundation: follow-up tracking is step 3 pointed at leads instead of keywords.

Open decisions and next steps

The one decision that unlocks everything else is who Vexa is for first.

- ☐ Pick the first user group (outreach for solo operators, or another specific group)
- ☐ Name one real person in that group and ask what they repeat every day
- ☐ Remove the accessibility-service code path from Vexa
- ☐ Switch email to the Gmail API and get step 1 of the build working
- ☐ Redesign memory as distilled facts with merge and expiry
- ☐ Write a plain-language privacy statement for the app
- ☐ If WhatsApp is in scope, check WhatsApp Business API access and costs